

MTH 202 Chapter 1: Three-Dimensional Space & Vectors

1.6 Planes in 3-Space Worksheet

1. Determine whether each of the following statements are true or false.

a) Two planes parallel to a third plane are parallel to each other.

b) Two lines perpendicular to a plane are parallel to each other.

c) Two planes that are parallel to a line are parallel.

d) Two planes that are perpendicular to a line are parallel.

e) A plane and a line either intersect or a parallel.

2. Find the equation of the plane.

a) The plane through the origin and perpendicular to the vector $\langle 1, -2, 5 \rangle$.

b) The plane through the point $(5, 3, 5)$ and with normal vector $2\mathbf{i} + \mathbf{j} - \mathbf{k}$.

c) The plane through the point $(2, 0, 1)$ and perpendicular to the line $x = 3t$, $y = 2 - t$, $z = 3 + 4t$.

d) The plane through the point $(1, -1, -1)$ and parallel to the plane $5x - y - z = 6$.

e) A plane through the point $(2, 4, 6)$ and perpendicular to the plane $z = x + y$.

f) The plane through the points $(0,1,1)$, $(1,0,1)$, & $(1,1,0)$.

g) Find an equation of the plane that contains the line $L = \langle 1+t, 1, 2t \rangle$ and the point $(2,4,0)$.

- h) Find an equation for the plane containing the point $(1, 2, 3)$ and the line L parameterized by $x = 1 + t$, $y = 2t$, & $z = 2 - t$.

- i) Find the equation of the plane P consisting of all points that are equidistance from $A = (2, 0, 1)$ and $B = (4, 2, 5)$.

3. Determine parametric equations for the line which goes through the point $(3, 1, 4)$ and is perpendicular to the plane $3x + 1y - 4z = 6$.

4. Suppose that two planes have (non-zero) normal vectors $\mathbf{n}_1$ and $\mathbf{n}_2$ respectively, and that $\mathbf{n}_1 \times \mathbf{n}_2 = \mathbf{0}$. Which of the following statements could possibly be true?
- A. The planes intersect in a line.
 - B. The planes are orthogonal to each other.
 - C. The planes are parallel to each other.
 - D. The planes are equal to each other.

5. Find the angle $0 \leq \theta \leq \frac{\pi}{2}$ between the planes $z = x + \sqrt{2}y$ and $x - z = 5$. Express your answer in radians.

6. Give a vector $\mathbf{v}$ perpendicular to the plane that contains the lines $x = 1 + t$, $y = 2 - 3t$, $z = 2 + 4t$ and $x = 2$, $y = -1 + 2t$, $z = 6 - t$.

7. Which statement is true in $\mathbb{R}^3$?

- a) Two planes perpendicular to a third plane are parallel.
- b) Two lines parallel to the same plane are parallel.
- c) Two lines either intersect or are parallel.
- d) Two planes either intersect or are parallel.

8. Consider the line L given by $x = 2 + 2t$, $y = 3$, $z = -1 + 2t$. Is L parallel to the plane $z - x - y = 2$?

9. Let A be the plane given by $x - z = 1$ and B the plane given by $x + y + z = 2$.

a) Find a normal vector $\mathbf{n}$ for the plane A .

b) Find the angle between the two planes.

c) Find the equation of a plane C which is perpendicular to both A and B .

10. Suppose that L is the line given by $x = 1 + 2t$, $y = -t$, $z = 2 + t$. Find the point P of the intersection of the line L with the plane whose equation is $3x - 2y + z = 14$.

11. Let P be the plane with equation $x + 2z = 0$. Find the distance from the point $(-1, 3, 0)$ to the plane P .
12. Determine the distance between the point $(-6, 3, 5)$ and the plane $x - 2y - 4z = 8$.
13. Find the distance between the two parallel planes $2x - 3y + z = 4$ and $4x - 6y + 2z = 3$.